"Ensemble Learning"

Wisdom of crowd: collective opinions of a group of indiviuals, when aggregated, is often more accurate than the opinion of any single even an expert.

Core Idea of Ensemble Learning:

Prediction stage

Ensemble learning algorithm → collection of algorithm

It all model same ← give different groups data to different model

It all different ← give train data full

give a sample to Predict. every model will predict a value and majority value will be final prediction

Types of Ensemble Learning:

4 types

- **Voting** — Base models are different algorithms
- **Bagging** — all same model
- **Boosting** — same models
- **Stacking** — train different models take prediction of all and make new data the 2nd stage use anothe algo for final prediction (models have weights)

Why ensemble works?

⇒ Take result of all model and build a more modified line/curve

Benefits:

- Improvement in performance
- Reduce bias and variance
- Robustness (changes in data doesn't reduce performance)

when to use Ensemble:

Always

Voting Ensemble:

Different base models (got full data to train)

Core Intuition:

- It's based on the idea of combining multiple diverse and independent models to make more accurate and robust prediction.

⇒ 2 main types:

1. Hard voting: Majority vote → final prediction is the class, most model agree on
2. Soft Voting: Average predicted probabilities. choos the class with highest mean Probability.

Why it works? → works based on principle: "Wisdom of Crowd"

☑ Error Reduction (Variance Reduction)

If individual models make uncorrelated errors, combining them helps cancel out mistakes. Specially helpful when model is overfit.

☑ Diversity Among Models:

Different algorithms or same algo trained with different hypermeters/bootstrapped data can capture different aspect. Leading better generalization.

☑ Law of Large Numbers:

when models are accurate more than 50% of the time aggreting them increase the chances of accuracy.

Assumptions of Voting:

(i) Model Diversity ⇒ models must make different types of errors (error should not be highly correlated).

(ii) Individual Base Learners are Better Than Random
One best is individually better than combining many bad one's.

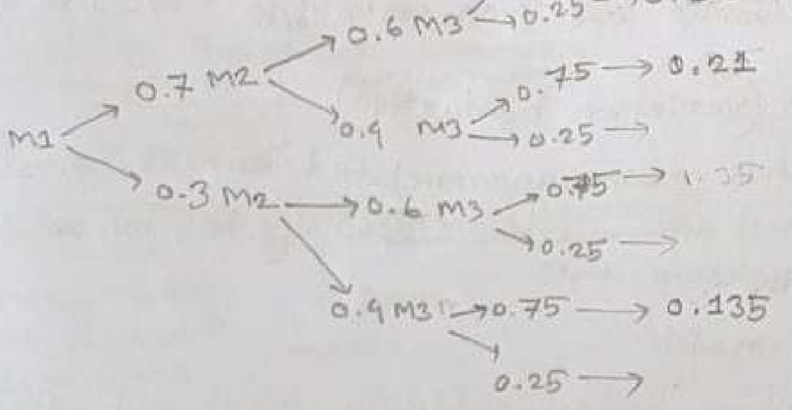
(iii) Indepence of Error:
errors should be uncorrelated across models. Same error on same data won't improve result.

(iv) Models must be independent in nature.

(v) Every models prediction must be greater than 50 % accurate.

at least 2 correct → are correct

M1	M2	M3
0.7	0.6	0.75
0.3	0.4	0.25



total
0.765

Bagging Ensemble: (Bootstrap + Aggregation)

- All models are same
- Every model gets a subset of train data.

Intuition

① Bootstrap Sampling: randomly takes n samples from train data and make n different train data set. Each sample subset (train set) must be same size.

② Model Training: train n models (all same) on new sample train data.

③ For prediction: get predicted value from each model and aggregate the prediction.

- For classification: majority vote.
- Regression: average of all value.

Why we use it?

- Reduce variance of unstable models (like Decision Tree)
- Prevent Overfitting
- Increase robustness

When to use it!

- working with high variance mode
  - (i) Decision Tree
  - (ii) KNN
- want to increase accuracy without increasing bias.
- enough rows to make samples
- want to make more stable model

Avoid Bagging when?

- already has low variance and high bias [Linear regression]
- little dataset

Note: without replacement one sample dataset D1 can have duplicate rows.

Types of Bagging:

① Standard Bagging
- multiple same model
- each model is trained on a bootstrapped dataset.
- combine the result.

② Random Subspace Method (Feature Bagging)
- each model gets a random subsets of features.
- helps to reduce feature correlation between models
- used when large numbers of features.

③ Random Forest
- use decision trees as base model
- Bootstrap sampling on rows
- Random subset selection of features at each split.

④ Pasting (without replacement)
- Like Standard bagging but no duplicate rows.
- used for large data

⑤ Bootstrap with weighted voting
- Some model get more importance on prediction.
- used when base model has different learning power.

⑥ Bagging with different base models
- less common
- used base model [KNN + SVM + Decision Tree]
- Basically mix Bagging and voting

Out of Bag Sample (OOB)

a performance estimator without using a seperate validation set or cross-validation.

while doing bootstraping there are some samples are left out of the model's trainning set. These are called out of Bag samples.

on average 63% of training data is included in each bootstrap sample and 37% OOB per model.

OOB is only available when bootstrap=True

Less stable than cross-validation when dataset is very small.

Bagging Tips:

- Bagging generally gives better result than pasting.
- Good result comes when max samples 25% to 50%.
- Random patches and subspaces should be used while dealing with high dimensional data
- use GridSearchCV to get best parameters.

"Random Forest"

Random Forest is a powerful ensemble learning algorithm used for both classification and Regression.

Random | Forest — why this name?

- Random → Bagging Based → Bootstrap → selected data randomly.
- Forest → group of trees

A bagging technique where we use Decision Tree as the base model.

Bootstrap sampling

- row sampling
- column/feature sampling

with replacement
- sample data may have duplicate value

without replacement
- sample data will be unique

why Random Forest works well?
cause random forest reduce the variance with increasing the bias.

But how?
cause RF make a group of full grown decision trees and as full grown so bias will be very very low.

* How it maintain low variance?
As RF make subsets of train data and train multiple DT. so if changes happens in train data then this changes splits and impacts are distributed so change in variance not happen

Difference Between Bagging and Random Forest

Feature	Bagging	Random Forest
Base model	Any	only Decision Tree
Feature selection at split	All feature	Random subsets of feature
Tree correlation	Higher	Low (feature random)
Bias	slightly lower	slightly higher
variance	Reduced	Reduced even more

Feature Importance:

a value we find that tell the importance level of each feature.

Tells how much each input feature contributes to the model's prediction.

why we use it?

(i) understadability: Helps to interpret what model is learning
(ii) Feature Selection: Remove unimportant features to reduce overfitting. and improve performance.
(iii) Dimensionality Reduction
(iv) Insight for domain experts: show what factors influence outcomes.

feature importance = total reduction in impurity the feature contributes across the whole tree.

How Decision tree calculate feature importance.

① At each node calculate impurity reduction and tree picks the best reduction

$$gini = 1 - \sum_{i=1}^{col} p_i^2$$

② Impurity decrease at a split:

$$decrease = I_{parent} - \left(\frac{N_{left}}{N} \cdot I_{left} + \frac{N_{right}}{N} I_{right} \right)$$

③ sum of all nodes impurity reduction (sum of threshold)

④ Normalize accross all feature.

Importance of i-th feature

$$Imp(i) = \frac{F_i}{\sum_{k=1}^{col} F_k}$$

For random forest with T trees. each tree gives a key value pair feature name with its importance for a tree.

$$Importance(j) = \frac{1}{T} \sum_{t=1}^{T} \frac{\text{Total impurity decrease from } j \text{ in } t}{\sum_{k=1}^{col} \text{Total impurity decrease for all feature in } t}$$

nomalize the importance = -

$$normImp(j) = \frac{Importance(j)}{\sum_{k} Importance(k)}$$